

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 Total Marks:50

Annexure A
CONCEPT MAPPING

Code of Conduct:

I _E. Lakshmi Narayanan_(192524249) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
E. Lakshmi Narayanan

Annexure A
CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat’s Little Theorem, Euler’s Phi Function and Euler’s Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify.

CLASSICAL ENCRYPTION

Substitution Techniques

Replace
Characters

Changes
Character
Values

Examples:
Caesar,
Vigenère

Transposition Techniques

Rearrange
Characters

Changes
Position
Only

Examples:
Rail Fence,
Columnar

Similarities

- Both Encrypt Plaintext
- Use Secret Keys

Substitution:
More Secure
(Frequency
Analysis)



Transposition:
Less Secure
(Pattern
Analysis)



Security Attacks Framework

❑ Passive Attack (Eavesdropping)

THREATS: Unauthorized Access
↓
RISKS: Data Leakage | Financial Loss

❑ Active Attack (Modification)

THREATS: Data Tampering | Malware
↓
RISKS: Incorrect Data | Data Corruption

❑ Physical Attack (Device Theft)

THREATS: Hardware Loss | System Damage
↓
RISKS: Service Failure | Downtime

↓ ALL ATTACKS CONVERGE TO ↓

SECURITY GOALS: Protecting systems from all attack vectors requires a multi-layered defense strategy combining proactive and reactive controls.

❑ Confidentiality

Controls: Encryption | Access Control
↓
Ensures only authorized users can access sensitive data and resources.

❑ Integrity

Controls: Hashing | Digital Signatures
↓
Ensures data accuracy and prevents unauthorized modification or tampering.

SUBSTITUTION CIPHERS

CAESAR CIPHER

Key: Shift - Simple - Weak

MONOALPHABETIC CIPHER

Key: 26 Letters - Moderate - Better

VIGENÈRE CIPHER

Key: Keyword - Complex - Stronger

HILL CIPHER

Key: Matrix - Most Complex - Strongest

MOST SECURE = HILL CIPHER (USES MATRIX OPERATIONS, DIFFICULT TO CRACK)

STEGANOGRAPHY

CONCEPTS

THE ART OF HIDDEN COMMUNICATION

Steganography conceals secret data within ordinary files, images, or messages — making hidden communication invisible to unintended recipients. Explore the three core pillars below.

DATA HIDING

Hide Secret Information inside ordinary digital files without detection.

Image & File Embedding conceals data inside pixels or file metadata seamlessly.

Resistance to Attacks ensures hidden data survives compression and manipulation.

WATERMARKING

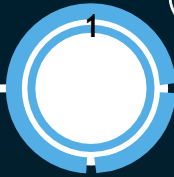
Ownership Protection embeds invisible identity markers into digital content.

Copyright Protection ensures creator rights are encoded and traceable.

SURVIVABILITY
Robust Hidden Information persists through transformations, edits, and noise —
converging to: **SECURE COMMUNICATION**

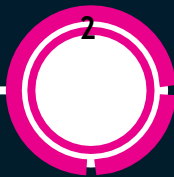
NUMBER THEORY

PUBLIC KEY CRYPTOGRAPHY (RSA) – ENCRYPTION & DECRYPTION



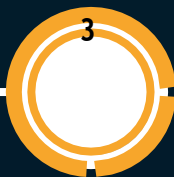
EXTENDED EUCLID ALGORITHM

Finds GCD of two integers using back-substitution. Foundation for computing modular inverses and key generation in cryptographic systems.



MODULAR INVERSE

Uses Extended Euclidean Algorithm to find the inverse. Essential for decryption key computation in RSA and modular arithmetic.



FERMAT'S LITTLE THEOREM

Simplifies Modular Arithmetic for prime moduli. States $a^{(p-1)} \equiv 1 \pmod{p}$, enabling fast exponentiation in cryptography.

